PARISHRAM 2.0 2026 (Class 12th)

Mathematics

DPP: 03

Relations and Functions

Q1 Let R be the relation in the set {1, 2, 3, 4} given by R = {(1, 2), (2, 2), (1, 1), (4, 4), (1, 3), (3, 3), (3, 2)). Choose the correct answer.

(A) R is reflexive and symmetric but not transitive

(B) R is reflexive and transitive but not symmetric

(C) R is symmetric and transitive but not reflexive

(D) R is an equivalence relation

Q2 Assertion (A) : The relation R in A = {1, 2, 3, 4, 5, 6} defined as R = {(x, y): y is divisible by x} is not an equivalence relation.

Reason (R) : The relation R will be an equivalence relation, if it is reflexive, symmetric and transitive.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Q3 The relation R defined in the set A = {1, 2, 3, 4, 5, 6, 7} by R = {(a, b) : both a and b are either odd or even}. Then, R is

(A) symmetric

(B) transitive

(C) an equivalence relation

(D) reflexive

Q4 If $\mathbf{R}$ be a relation on $\mathbf{A}$ such that $\mathbf{R}=\{(2,2),(3,3),(2,3),(3,2),(3,1),(2,1)\}$ on the set $\mathbf{A}=\{1,2,3\}$, then R is

(A) reflexive (B) symmetric

(C) equivalence (D) transitive

Q5 What type of a relation is "less than" in the set of real numbers?

(A) only symmetric

(B) only transitive

(C) only reflexive

(D) equivalence relation

Q6 Assertion (A) : Let L be a set of lines in a plane. R is a relation on L defined as $\mathbf{R}=\{(\mathbf{L}_1,\mathbf{L}_2):\mathbf{L}_1$ is parallel to $\mathbf{L}_2\}$. Then R is an equivalence relation.

Reason (R) : R is a transitive relation.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true

Q7 A relation R in set A = {1, 2, 3} is defined as R = {(1,1),(1,2),(2,2),(3,3)}. Which of the following ordered pair in R shall be removed to make it an equivalence relation in A?

(A) (1,1) (B) (1,2)

(C) (2,2) (D) (3,3)

Answer Key

Q1	B
Q2	A
Q3	C
Q4	D
Q5	B
Q6	D
Q7	B

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

R = {(1, 2), (2, 2), (1, 1), (4, 4), (1, 3), (3, 3), (3, 2))

It is seen that (a, a) ∈ R, for every a ∈ {1, 2, 3, 4}.

∴R is reflexive.

It is seen that (1, 2) ∈ R, but (2, 1) ∉ R.

∴R is not symmetric.

Also, it is observed that (a, b), (b, c) ∈ R ⇒ (a, c) ∈ R for all a, b, c ∈ {1, 2, 3, 4}.

∴ R is transitive.

Hence, R is reflexive and transitive but not symmetric.

Video Solution:

Q2 Text Solution:

Here, R = {(x, y): y is divisible by x} is a relation in the set A = {1, 2, 3, 4, 5, 6}.

Reflexive:

We know that x is divisible by x, which is true for all $\mathbf{x} \in \mathbf{A}$ i.e., $(\mathbf{x}, \mathbf{x}) \in \mathbf{R}$ for all $\mathbf{x} \in \mathbf{A}$.

So, R is reflexive.

Symmetric:

We observe that 6 is divisible by 2 . This means that $(2, 6) \in \mathbf{R}$ but $(6, 2) \notin \mathbf{R}$ So, R is not symmetric.

Transitive:

Let $(\mathbf{x}, \mathbf{y}) \in \mathbf{R}$ and $(\mathbf{y}, \mathbf{z}) \in \mathbf{R}$

Here y is divisible by x and z is divisible by y.

thus z is divisible by x. It means $(\mathbf{x}, \mathbf{y}) \in \mathbf{R}$.

For example, 2 is divisible by 1 and 4 is divisible by 2.

So, 4 is divisible by 1 . So, R is transitive

Thus, both (A) and (R) are true and (R) is the correct explanation of (A).

Video Solution:

Q3 Text Solution:

Given, any element a in A, both a and a must be either odd or even, so that $(\mathbf{a}, \mathbf{a}) \in \mathbf{R}$.

Further, $(\mathbf{a}, \mathbf{b}) \in \mathbf{R} \Rightarrow$ both a and b must be either odd or even $\Rightarrow (\mathbf{b}, \mathbf{a}) \in \mathbf{R}$.

Similarly, $(\mathbf{a}, \mathbf{b}) \in \mathbf{R}$ and $(\mathbf{b}, \mathbf{c}) \in \mathbf{R} \Rightarrow$ all elements a, b, c must be either even or odd simultaneously $\Rightarrow (\mathbf{a}, \mathbf{c}) \in \mathbf{R}$. Hence, R is an equivalence relation.

Video Solution:

Q4 Text Solution:

We have,

$\mathbf{R} = \{(2, 2), (3, 3), (2, 3), (3, 2), (3, 1), (2, 1)\}$

Reflexive:

Since $(1, 1) \notin \mathbf{R}$, therefore R is not reflexive

Symmetric:

Since $(3, 1) \in \mathbf{R}$, but $(1, 3) \notin \mathbf{R}$, therefore R is not symmetric.

Transitive:

Since, for every element $(\mathbf{a}, \mathbf{b}) \in \mathbf{R}$ and $(\mathbf{b}, \mathbf{c}) \in \mathbf{R} \Rightarrow (\mathbf{a}, \mathbf{c}) \in \mathbf{R}$, therefore R is transitive.

Video Solution:

Q5 **Text Solution:**

Let A be the set of all real numbers, and R be the relation 'less than' i.e., < on A, then R is

Reflexive:

The relation R is not reflexive, since a is not less than for any natural number a.

Symmetric:

The relation R is not symmetric, since if (a, b) ∈ R , then a is less than b but b is not less than a, i.e., (b,a) $\notin \mathbf{R}$

Transitive:

The relation R is transitive, since $\mathbf{a} < \mathbf{b}$ and $\mathbf{b} < \mathbf{c} \Rightarrow \mathbf{a} < \mathbf{c}$ $\mathbf{i.e.,}\ (\mathbf{a}, \mathbf{b}) \in \mathbf{R},\ (\mathbf{b}, \mathbf{c}) \in \mathbf{R} \Rightarrow (\mathbf{a}, \mathbf{c}) \in \mathbf{R}.$

Video Solution:

Q6 **Text Solution:**

Reflexive:

Since a line is not parallel to itself. Therefore, relation R is not a reflexive relation.

Symmetric:

A relation R on a set L is said to be symmetric relation iff $(\mathbf{a}, \mathbf{b}) \in \mathbf{R} \Rightarrow (\mathbf{b}, \mathbf{a}) \in \mathbf{R}$ for all $\mathbf{a}, \mathbf{b} \in \mathbf{L}.\ \mathbf{If}\ \mathbf{L_1} \| \mathbf{L_2} \Rightarrow \mathbf{L_2} \| \mathbf{L_2}$. Thus, given relation is also symmetric.

Transitive:

A relation R on a set L is said to be transitive relation iff $(\mathbf{a}, \mathbf{b}) \in \mathbf{R}, (\mathbf{b}, \mathbf{c}) \in \mathbf{R} \Rightarrow (\mathbf{a}, \mathbf{c}) \in \mathbf{R}$ for all $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbf{L}.\ \mathbf{If}\ \mathbf{L_1} \| \mathbf{L_2}$. Therefore, given $\mathbf{and}\ \mathbf{L_2} \| \mathbf{L_3} \Rightarrow \mathbf{L_1} \| \mathbf{L_3}$ relation is transitive relation.

Relation which is reflexive, symmetric and transitive is called equivalence relation.

Thus, (A) is false but (R) is true.

Video Solution:

Q7 **Text Solution:**

We have, $\mathbf{A} = \{1, 2, 3\}$ and $\mathbf{R} = \{(1, 1), (1, 2), (2, 2), (3, 3)\}$

Here we have $(1, 2)$, so we need to have $(2, 1)$ to make it symmetric. But if we remove $(1, 2)$, this relation will be symmetric, reflexive and transitive i.e. equivalence relation.

So, if we remove $(1, 2)$ then R becomes equivalence relation.

Video Solution: